

Original Article

An Efficient Ranked Multi-Keyword Search for Multiple Data Owners over Encrypted Cloud Data

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Abstract – Cloud computing enables users to outsource data storage and access services to remote servers. However, ensuring data privacy while supporting efficient search over encrypted data remains a significant challenge. Existing solutions mainly focus on single-owner models, whereas real-world cloud systems often involve multiple data owners. This paper presents a Privacy-Preserving Ranked Multi-keyword Search scheme in a Multi-owner Model (PRMSM). The proposed approach allows secure searching without revealing keyword or data information to the cloud server. A novel additive order and privacy-preserving function is introduced to rank search results while protecting relevance scores. Additionally, dynamic key generation and user authentication mechanisms are implemented to enhance security and prevent unauthorized access. Experimental evaluation demonstrates that the proposed system achieves improved efficiency and security compared to existing approaches.

Keywords – Cloud Computing, Searchable Encryption, Multi-keyword Search, Privacy Preservation, Data Security.

1. INTRODUCTION:

Cloud storage systems provide scalable and cost-effective data management services. However, outsourcing sensitive data to third-party cloud providers raises serious privacy concerns. Encryption techniques can protect data confidentiality, but they limit usability, particularly for keyword-based search operations.

Traditional searchable encryption schemes support secure keyword search but are primarily designed for single data owners. In practice, cloud environments involve multiple data owners who independently upload encrypted data. This scenario introduces new challenges in maintaining privacy, efficiency, and usability.

This paper proposes a secure multi-owner searchable encryption scheme that supports ranked multi-keyword queries. The system enables cloud servers to process search requests without accessing plaintext data or keywords. Furthermore, it ensures that only authorized users can perform searches

while maintaining data confidentiality.

2. RELATED WORK:

Searchable encryption was first introduced by Song et al., enabling keyword-based searches over encrypted data. Subsequent research expanded this concept to cloud environments, improving efficiency and functionality.

Wang et al. proposed ranked keyword search over encrypted data, enhancing search relevance. Cao et al. introduced multi-keyword ranked search schemes but limited them to single-owner models. Other works explored fuzzy keyword search and attribute-based search mechanisms to improve usability.

Despite these advancements, most existing approaches fail to address multi-owner scenarios effectively. Additionally, challenges such as secure ranking, user authentication, and efficient key management remain unresolved.

3. PROBLEM STATEMENTS:

Sensitive data owners are reluctant to outsource their data to cloud servers without strong guarantees of confidentiality and query privacy. Existing systems either lack support for multiple data owners or fail to provide efficient ranked search functionality while preserving privacy.

4. PROPOSED SYSTEM:

The proposed PRMSM system introduces a secure and efficient framework for multi-owner cloud environments.

Key Features:

- Supports multi-keyword ranked search over encrypted data.
- Enables multiple data owners to encrypt data using independent keys.
- Allows authorized users to perform secure searches without accessing secret keys.
- Introduces an Additive Order and Privacy-Preserving Function (AOPPF) for secure ranking.
- Implements dynamic key generation and authentication protocols.
- Supports user revocation and system scalability.

Security Mechanisms:

- AES algorithm for symmetric encryption
- RSA algorithm for secure key generation
- Digital Signature Standard (DSS) for authentication

5. SYSTEM ARCHITECTURE:

The system consists of four main entities:

- Data Owners: Upload encrypted data and keywords
- Cloud Server: Stores data and performs secure search operations
- Data Users: Submit search queries and retrieve results
- Authentication Authority: Manages keys and user verification

The architecture ensures that sensitive information is never exposed during search operations.

6. ALGORITHMS USED:

- A. AES Algorithm: A symmetric encryption technique based on substitution-permutation operations, ensuring strong data confidentiality.
- B. RSA Algorithm: An asymmetric cryptographic method used for secure key exchange and encryption.
- C. Secure Key Distribution (DSS): Ensures safe transmission of keys using authenticated communication channels such as email or messaging systems.

7. PERFORMANCE ANALYSIS:

The proposed PRMSM scheme is evaluated against existing methods such as SRMSM and MRSE.

Results indicate that PRMSM provides:

- Better efficiency in multi-owner environments
- Enhanced security against unauthorized access
- Improved ranking accuracy without compromising privacy

8. ADVANTAGES AND LIMITATIONS:

Advantages:

- Supports multiple data owners
- Enables efficient multi-keyword search
- Preserves data and query privacy
- Provides secure user authentication and revocation

Limitations:

- Computational overhead due to encryption operations
- Complexity in key management

9. CONCLUSION:

This paper presents a privacy-preserving multi-keyword ranked search system designed for multi-owner cloud environments. The proposed PRMSM framework addresses limitations of existing systems by enabling secure, efficient, and scalable data retrieval. By integrating advanced cryptographic techniques and secure ranking mechanisms, the system ensures strong privacy protection while maintaining usability.

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